

Jacob Pieniazek

Lead Data Scientist | Causal ML & Experimentation

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EXPERIENCE

84.51° | RETAIL DATA SCIENCE | LEAD DATA SCIENTIST

Feb 2026 – Present | Cincinnati, OH

- Serve as technical lead for four data scientists delivering a production causal measurement platform supporting 1,000+ annual enterprise campaigns and measurement of \$100M+ in incremental revenue.
- Lead a two-scientist causal ML R&D team whose CATE targeting delivers roughly twice the average incremental gain of standard targeting across multiple production and validated use cases.
- Served as causal inference expert for a third-party profit-sharing partnership; designed an experimental audit revealing material positive bias in reported incrementality and preventing significant overpayment.
- Lead agentic-AI adoption across development, CI automation, and end-user reporting.
- Lead the causal inference SIG and internal training program, facilitating biweekly learning and piloting curricula in econometrics, experimental design, and causal ML.

Tech: *Kedro, Ibis, EconML, MLflow, LiteLLM, LangChain, Claude Code, Docker, GitHub Actions*

84.51° | SENIOR DATA SCIENTIST

Jul 2024 – Feb 2026 | Cincinnati, OH

- Architected an extensible, interoperable measurement platform with Kedro and Ibis; led a full-stack migration from Azure Data Factory to GitHub Actions and Databricks, cutting pipeline costs by 70% through workflow and Spark optimizations.
- Designed a multi-campaign experiment framework that captures cumulative and interaction effects while preserving exogenous variation, empowering the business to understand initiative effects at scale beyond bespoke marketing campaigns.

Tech: *Python, PySpark, Databricks, Kedro, Ibis*

84.51° | DATA SCIENTIST

Jun 2022 – Jul 2024 | Cincinnati, OH

- Owned an enterprise uplift pipeline and CI/CD lifecycle supporting year-round campaign measurement, establishing engineering practices for its replacement.

Tech: *Python, PySpark, Azure Data Factory, Azure DevOps*

PROJECTS

CAML | CAUSAL AUTOML FRAMEWORK (DOCS) (CODE)

- Redesigning an open-source AutoML framework for causal learning and CATE estimation, including model selection, out-of-sample validation, and policy prescription.

Tech: *Python, EconML, DoubleML, FLAML, Ray, Spark*

NETBEAT | NETWORK DIAGNOSTICS CLI (CRATES) (CODE)

- Built a lightweight client/server CLI for network throughput testing, monitoring, and diagnostics.

Tech: *Rust*

MORE | LITELLM PULSE, TECHNICAL ARTICLES & TALKS (PROJECTS) (ARTICLES)

- Additional work spanning LLM observability, causal ML, econometrics, optimization, and statistical learning.

Tech: *Python, FastAPI, SQLite, Docker*

SKILLS

CORE STRENGTHS

- Causal & Econometric Methods
- Experimentation & Causal ML
- AI/ML & Agentic Systems
- Software & Package Engineering
- Technical Leadership & Delivery

PROGRAMMING

Python • SQL • Rust • Bash
R • Stata • LaTeX

AI / AGENTS

LangChain • RAG • LiteLLM
MCP / ACP • Agentic Coding
OpenCode • Claude Code

ML / CAUSAL

EconML • DoubleML • DoWhy
scikit-learn • XGBoost • FLAML
PyTorch • Hugging Face

DEVOPS / SYSTEMS

Linux • Git • Docker
Kubernetes • Helm
GitHub Actions • CI/CD

PYTHON / DATA TOOLS

Kedro • Ibis • uv • pytest
Databricks • Spark • Delta
Azure • MLflow

EDUCATION

MIAMI UNIVERSITY

M.A. ECONOMICS | GPA: 4.0
2022 | Full Graduate Assistantship

UNIVERSITY OF DAYTON

B.S. BUSINESS ECONOMICS &
APPLIED MATHEMATICAL ECONOMICS
Double Major | 2021 | GPA: 3.92
Summa Cum Laude, Honors Diploma
E. B. O'Leary Award of Excellence to the
Outstanding Senior Majoring in Economics

INTERESTS/HOBBIES

Open-Source Development •
Self-Hosting / Homelab • Home
Automation • Hiking / Camping
• Guitar • Audiophile Pursuits